Student Management System (Attenda)

System Architecture Specification & Technical Documentation — Category: Full Stack Academic Administration

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Repository: https://github.com/rakshapatil526/Student-Management-System

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1. Overview & System Purpose

Attenda is an institutional Student Management System designed for academic colleges and universities. Built with Python Django, HTML5/CSS3/JavaScript, and SQLite, it allows department heads and administrative staff to manage enrollments, track CGPA academic performance metrics, manage departments and degree courses, and generate official student records.

2. Core Administrative Modules

• **Executive Dashboard:** Live metric counters (Total Students, Academic Departments, Degree Courses, Institutional Average CGPA) and Chart.js distribution charts.

• **Enrollment Workflow:** Multi-section student registration with cascading dynamic dropdowns for departments and courses, and client-side photo previews.

• **Directory & Filtering:** Paginated data grid with multi-criteria search (Name, Roll Number, Department) and one-click CSV export.

• **Student Academic Profile:** Detailed scorecard view displaying semester-wise performance tier (Distinction, First Class, Second Class, Remedial).

3. Architecture & Data Integrity

• **Backend:** Django Class-Based Views (CBVs), Django ORM queries with prefetch optimizations, and server-side form validations.

• **Database:** Relational tables for Departments, Courses, Students, and Semester Records with foreign-key cascades.

• **Distinction Note:** This project is a full-stack academic record management system. It is distinct from the machine learning Student Performance Prediction model.